

Mark Scheme

Q1.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$\frac{1}{\sqrt{4-x}} = (4-x)^{-\frac{1}{2}} = 4^{-\frac{1}{2}} \left(1 - \frac{x}{4}\right)^{-\frac{1}{2}}$	M1	This mark is given for rearranging $\frac{1}{\sqrt{4-x}}$ to attempt a binomial expansion
	$\left(1 - \frac{x}{4}\right)^{-\frac{1}{2}} =$	M1	This mark is given for an attempt at a binomial expansion
	$1 + \left(-\frac{1}{2}\right)\left(-\frac{x}{4}\right) + \frac{\left(-\frac{1}{2}\right) \times \left(-\frac{3}{2}\right)}{2} \left(-\frac{x}{4}\right)$	A1	This mark is given for a fully correct binomial expansion
	$\frac{1}{\sqrt{4-x}} = \frac{1}{2} + \frac{1}{16}x + \frac{3}{256}x^2$	A1	This mark is given for a fully correct expansion with the first three terms
(b)(i)	$x = -14$, since the expansion is only valid for $ x < 4$	B1	This mark is given for the correct value chosen with a correct reason
(b)(ii)	$x = -\frac{1}{2}$, since the smaller value will give the more accurate approximation	B1	This mark is given for the correct value chosen with a correct reason

Q2.

Question	Scheme	Marks	AOs
	${}^7C_4 a^3 (2x)^4$	M1	1.1b
	$\frac{7!}{4!3!} a^3 \times 2^4 = 15120 \Rightarrow a = \dots$	dM1	2.1
	$a = 3$	A1	1.1b
		(3)	

(3 marks)

Notes:

M1: For an attempt at the correct coefficient of x^4 .

The coefficient must have

- the correct binomial coefficient
- the correct power of a
- 2 or 2^4 (may be implied)

May be seen within a full or partial expansion.

Accept ${}^7C_4 a^3 (2x)^4$, $\frac{7!}{4!3!} a^3 (2x)^4$, $\left(\frac{7}{4}\right) a^3 (2x)^4$, $35a^3 (2x)^4$, $560a^3 x^4$, $\left(\frac{7}{4}\right) a^3 16x^4$ etc.

or ${}^7C_4 a^3 2^4$, $\frac{7!}{4!3!} a^3 2^4$, $\left(\frac{7}{4}\right) a^3 2^4$, $35a^3 2^4$, $560a^3$ etc.

or ${}^7C_3 a^3 (2x)^4$, $\frac{7!}{4!3!} a^3 (2x)^4$, $\left(\frac{7}{3}\right) a^3 (2x)^4$, $35a^3 (2x)^4$, $560a^3 x^4$, $\left(\frac{7}{3}\right) a^3 16x^4$ etc.

or ${}^7C_3 a^3 2^4$, $\frac{7!}{4!3!} a^3 2^4$, $\left(\frac{7}{3}\right) a^3 2^4$, $35a^3 2^4$, $560a^3$

You can condone missing brackets around the "2x" so allow e.g. $\frac{7!}{4!3!} a^3 2x^4$

An alternative is to attempt to expand $a^7 \left(1 + \frac{2x}{a}\right)^7$ to give $a^7 \left(\dots \frac{7 \times 6 \times 5 \times 4}{4!} \left(\frac{2x}{a}\right)^4 \dots\right)$

Allow M1 for e.g. $a^7 \left(\dots \frac{7 \times 6 \times 5 \times 4}{4!} \left(\frac{2x}{a}\right)^4 \dots\right)$, $a^7 \left(\dots \left(\frac{7}{4}\right) \left(\frac{2x}{a}\right)^4 \dots\right)$, $a^7 \left(\dots 35 \left(\frac{2x}{a}\right)^4 \dots\right)$ etc.

but condone missing brackets around the $\frac{2x}{a}$

Note that 7C_3 , $\left(\frac{7}{3}\right)$ etc. are equivalent to 7C_4 , $\left(\frac{7}{4}\right)$ etc. and are equally acceptable.

If the candidate attempts $(a + 2x)(a + 2x)(a + 2x) \dots$ etc. then it must be a complete method to reach the required term. Send to review if necessary.

dM1: For " 560 " $a^3 = 15120 \Rightarrow a = \dots$ Condone slips on copying the 15120 but their " 560 " must be an attempt at

${}^7C_4 \times 2$ or ${}^7C_4 \times 2^4$ and must be attempting the cube root of $\frac{15120}{560}$. **Depends on the first mark.**

A1: $a = 3$ and no other values i.e. ± 3 scores A0

Note that this is fairly common:

$${}^7C_4 a^3 2x^4 = 70a^3 x^4 \Rightarrow 70a^3 = 15120 \Rightarrow a^3 = 216 \Rightarrow a = 6$$

and scores **M1 dM1 A0**

Q3.

Question	Scheme	Marks	AOs
(a)	$\sqrt{4-9x} = 2(1 \pm \dots)^{\frac{1}{2}}$	B1	1.1b
	$\left(1 - \frac{9x}{4}\right)^{\frac{1}{2}} = \dots + \frac{\frac{1}{2} \times \left(-\frac{1}{2}\right) \left(-\frac{9x}{4}\right)^2}{2!} \text{ or }$ $\dots + \frac{\frac{1}{2} \times \left(-\frac{1}{2}\right) \times \left(-\frac{3}{2}\right) \left(-\frac{9x}{4}\right)^3}{3!}$	M1	1.1b
	$1 + \frac{1}{2} \times \left(-\frac{9x}{4}\right) + \frac{\frac{1}{2} \times \left(-\frac{1}{2}\right) \left(-\frac{9x}{4}\right)^2}{2!} + \frac{\frac{1}{2} \times \left(-\frac{1}{2}\right) \times \left(-\frac{3}{2}\right) \left(-\frac{9x}{4}\right)^3}{3!}$	A1	1.1b
	$\sqrt{4-9x} = 2 - \frac{9x}{4} - \frac{81x^2}{64} - \frac{729x^3}{512}$	A1	1.1b
	(4)		
(b)	States that the approximation will be an <u>overestimate</u> since all terms (after the first one) in the expansion are negative (since $x > 0$)	B1	3.2b
		(1)	
(5 marks)			
Notes:			

(a)

B1: Takes out a factor of 4 and writes $\sqrt{4-9x} = 2(1 \pm \dots)^{\frac{1}{2}}$ or $\sqrt{4}(1 \pm \dots)^{\frac{1}{2}}$ or $4^{\frac{1}{2}}(1 \pm \dots)^{\frac{1}{2}}$

M1: For an attempt at the binomial expansion of $(1+ax)^{\frac{1}{2}}$ $a \neq 1$ to form term 3 or term 4 with the correct structure. Look for the correct binomial coefficient multiplied by the corresponding power of x e.g.

$$\frac{\left(\frac{1}{2}\right)\left(\frac{1}{2}-1\right)}{2!}(\dots x)^2 \text{ or } \frac{\left(\frac{1}{2}\right)\left(\frac{1}{2}-1\right)\left(\frac{1}{2}-2\right)}{3!}(\dots x)^3 \text{ where } \dots \neq 1$$

Condone missing or incorrect brackets around the x terms but the binomial coefficients must be correct. Allow $2!$ and/or $3!$ or 2 and/or 6 . Ignore attempts to find more terms.

Do not allow notation such as $\left(\frac{1}{2}\right)$, $\left(\frac{1}{2}\right)$ unless these are interpreted correctly.

A1: Correct expression for the expansion of $\left(1 - \frac{9x}{4}\right)^{\frac{1}{2}}$ e.g.

$$1 + \frac{1}{2} \times \left(-\frac{9x}{4}\right) + \frac{\frac{1}{2} \times \left(\frac{1}{2}-1\right) \left(-\frac{9x}{4}\right)^2}{2!} + \frac{\frac{1}{2} \times \left(\frac{1}{2}-1\right) \times \left(\frac{1}{2}-2\right) \left(-\frac{9x}{4}\right)^3}{3!}$$

which may be left unsimplified as shown but the bracketing must be correct unless any missing brackets are implied by subsequent work. If the 2 outside this expansion is only partially applied to this expansion then score A0 but if it is applied to all terms this A1 can be implied.

OR at least 2 correct simplified terms for the final expansion from, $-\frac{9x}{4}$, $-\frac{81x^2}{64}$, $-\frac{729x^3}{512}$

A1: $\sqrt{4-9x} = 2 - \frac{9x}{4} - \frac{81x^2}{64} - \frac{729x^3}{512}$ oe and condone e.g. $2 + \frac{-9x}{4} - \frac{81x^2}{64} - \frac{729x^3}{512}$

Allow equivalent mixed numbers and/or decimals for the coefficients e.g.:

$$\left(\frac{9}{4}, 2\frac{1}{4}, 2.25\right), \left(\frac{81}{64}, 1\frac{17}{64}, 1.265625\right), \left(\frac{729}{512}, 1\frac{217}{512}, 1.423828125\right)$$

Ignore any extra terms if found. Allow terms to be "listed" and apply isw once a correct expansion is seen. Allow recovery if applicable e.g. if an "x" is lost then "reappears".

Direct expansion in (a) can be marked in a similar way:

$$\sqrt{4-9x} = (4-9x)^{\frac{1}{2}} = 4^{\frac{1}{2}} + \left(\frac{1}{2}\right)4^{-\frac{1}{2}} \times (-9x)^1 + \left(\frac{1}{2}\right)\left(\frac{1}{2}-1\right)4^{-\frac{3}{2}} \times \frac{(-9x)^2}{2!} + \left(\frac{1}{2}\right)\left(\frac{1}{2}-1\right)\left(\frac{1}{2}-2\right)4^{-\frac{5}{2}} \times \frac{(-9x)^3}{3!}$$

B1: For 2 or $\sqrt{4}$ or $4^{\frac{1}{2}}$ as the constant term in the expansion.

M1: Correct form for term 3 or term 4.

E.g. $\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right) \times \frac{(-9x)^2}{2!}$ or $\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right) \times \frac{(-9x)^3}{3!}$ where $\dots \neq 1$

Condone missing brackets around the x terms but the binomial coefficients must be correct.

Allow 2! and/or 3! or 2 and/or 6. Ignore attempts to find more terms.

Do not allow notation such as $\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$, $\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ unless these are interpreted correctly.

A1: Correct expansion (unsimplified as above)

OR at least 2 correct simplified terms from, $-\frac{9x}{4}$, $-\frac{81x^2}{64}$, $-\frac{729x^3}{512}$

A1: $\sqrt{4-9x} = 2 - \frac{9x}{4} - \frac{81x^2}{64} - \frac{729x^3}{512}$ or and condone e.g. $2 + \frac{-9x}{4} - \frac{81x^2}{64} - \frac{729x^3}{512}$

Allow equivalent mixed numbers and/or decimals for the coefficients e.g.:

$$\left(\frac{9}{4}, 2\frac{1}{4}, 2.25\right), \left(\frac{81}{64}, 1\frac{17}{64}, 1.265625\right), \left(\frac{729}{512}, 1\frac{217}{512}, 1.423828125\right)$$

Ignore any extra terms if found. Allow terms to be "listed" and apply isw once a correct expansion is seen. Allow recovery if applicable e.g. if an "x" is lost then "reappears".

(b)

B1: States that the approximation will be an overestimate due to the fact that all terms (after the first one) in the expansion are negative or equivalent statements e.g.

- Overestimate because the terms are negative
- Overestimate as the terms are being taken away (from 2)

Condone "overestimate as every term is negative"

If you think a response is worthy of credit but are unsure then use Review.

This mark depends on having obtained an expansion in (a) of the form

$k - px - qx^2 - rx^3$ $k, p, q, r > 0$ but note that if e.g. one of the algebraic terms is zero or was "lost" or there are extra negative terms this mark is still available.

Q4.

Question	Scheme	Marks	AOs
(a)(i)	$50x^2 + 38x + 9 \equiv A(5x+2)(1-2x) + B(1-2x) + C(5x+2)^2$ $\Rightarrow B = \dots$ or $C = \dots$	M1	1.1b
	$B = 1$ and $C = 2$	A1	1.1b
(a)(ii)	E.g. $x = 0 \Rightarrow 9 = 2A + B + 4C$ $\Rightarrow 9 = 2A + 1 + 8 \Rightarrow A = \dots$	M1	2.1
	$A = 0^*$	A1*	1.1b
		(4)	
(b)(i)	$\frac{1}{(5x+2)^2} = (5x+2)^{-2} = 2^{-2} \left(1 + \frac{5}{2}x\right)^{-2}$ or $(5x+2)^{-2} = 2^{-2} + \dots$	M1	3.1a
	$\left(1 + \frac{5}{2}x\right)^{-2} = 1 - 2\left(\frac{5}{2}x\right) + \frac{-2(-2-1)}{2!}\left(\frac{5}{2}x\right)^2 + \dots$	M1	1.1b
	$2^{-2} \left(1 + \frac{5}{2}x\right)^{-2} = \frac{1}{4} - \frac{5}{4}x + \frac{75}{16}x^2 + \dots$	A1	1.1b
	$\frac{1}{(1-2x)} = (1-2x)^{-1} = 1 + 2x + \frac{-1(-1-1)}{2!}(2x)^2 + \dots$	M1	1.1b
	$\frac{1}{(5x+2)^2} + \frac{2}{1-2x} = \frac{1}{4} - \frac{5}{4}x + \frac{75}{16}x^2 + \dots + 2 + 4x + 8x^2 + \dots$	dM1	2.1
	$= \frac{9}{4} + \frac{11}{4}x + \frac{203}{16}x^2 + \dots$	A1	1.1b
(b)(ii)	$ x < \frac{2}{5}$	B1	2.2a
		(7)	
(11 marks)			
Notes			

(a)(i)

M1: Uses a correct identity and makes progress using an appropriate strategy (e.g. sub $x = \frac{1}{2}$) to find a value for B or C . May be implied by one correct value (cover up rule).

A1: Both values correct

(a)(ii)

M1: Uses an appropriate method to establish an equation connecting A with B and/or C and uses their values of B and/or C to find a suitable equation in A .

Amongst many different methods are:

Compare terms in $x^2 \Rightarrow 50 = -10A + 25C$ which would be implied by $50 = -10A + 25 \times 2$

Compare constant terms or substitute $x = 0 \Rightarrow 9 = 2A + B + 4C$ implied by $9 = 2A + 1 + 4 \times 2$

A1*: Fully correct proof with no errors.

Note: The second part is a proof so it is important that a suitable proof/show that is seen.

Candidates who write down 3 equations followed by three answers (with no working) will score M1 A1 M0 A0

(b)(i)

M1: Applies the key steps of writing $\frac{1}{(5x+2)^2}$ as $(5x+2)^{-2}$ and takes out a factor of 2^{-2} to form an

expression of the form $(5x+2)^{-2} = 2^{-2}(1+*x)^{-2}$ where * is not 1 or 5

Alternatively uses direct expansion to obtain $2^{-2} + \dots$

M1: Correct attempt at the binomial expansion of $(1+*x)^{-2}$ up to the term in x^2

Look for $1 + (-2)*x + \frac{(-2)(-3)}{2} *x^2$ where * is not 5 or 1.

Condone sign slips and lack of $*^2$ on term 3.

Alt Look for correct structure for 2nd and 3rd terms by direct expansion. See below

A1: For a fully correct expansion of $(2+5x)^{-2}$ which may be unsimplified. This may have been combined with their 'B'

A direct expansion would look like $(2+5x)^{-2} = 2^{-2} + (-2)2^{-3} \times 5x + \frac{(-2)(-3)}{2} 2^{-4} \times (5x)^2$

M1: Correct attempt at the binomial expansion of $(1-2x)^{-1}$

Look for $1 + (-1)*x + \frac{(-1)(-2)}{2} *x^2$ where * is not 1

dM1: Fully correct strategy that is dependent on the previous TWO method marks.

There must be some attempt to use their values of B and C

A1: Correct expression or correct values for p, q and r.

(b)(ii)

B1: Correct range. Allow also other forms, for example $-\frac{2}{5} < x < \frac{2}{5}$ or $x \in \left(-\frac{2}{5}, \frac{2}{5}\right)$

Do not allow multiple answers here. The correct answer must be chosen if two answers are offered

Q5.

Question	Scheme	Marks	AOs
(a)	$(1+8x)^{\frac{1}{2}} = 1 + \frac{1}{2} \times 8x + \frac{\frac{1}{2} \times -\frac{1}{2}}{2!} \times (8x)^2 + \frac{\frac{1}{2} \times -\frac{1}{2} \times -\frac{3}{2}}{3!} \times (8x)^3$	M1 A1	1.1b 1.1b
	$= 1 + 4x - 8x^2 + 32x^3 + \dots$	A1	1.1b
		(3)	
(b)	Substitutes $x = \frac{1}{32}$ into $(1+8x)^{\frac{1}{2}}$ to give $\frac{\sqrt{5}}{2}$	M1	1.1b
	Explains that $x = \frac{1}{32}$ is substituted into $1 + 4x - 8x^2 + 32x^3$ and you multiply the result by 2	A1ft	2.4
		(2)	
(5 marks)			
Notes:			



(a)

M1: Attempts the binomial expansion with $n = \frac{1}{2}$ and obtains the correct structure for term 3 or term 4.

Award for the correct coefficient with the correct power of x . Do not accept nC_r notation for coefficients.

For example look for term 3 in the form $\frac{\frac{1}{2} \times -\frac{1}{2}}{2!} \times (*x)^2$ or $\frac{\frac{1}{2} \times -\frac{1}{2} \times -\frac{3}{2}}{3!} \times (*x)^3$

A1: Correct (unsimplified) expression. May be implied by correct simplified expression

A1: $1 + 4x - 8x^2 + 32x^3$

Award if there are extra terms (even if incorrect).

Award if the terms are listed $1, 4x, -8x^2, 32x^3$

(b)

M1: Score for substituting $x = \frac{1}{32}$ into $(1+8x)^{\frac{1}{2}}$ to obtain $\frac{\sqrt{5}}{2}$ or equivalent such as $\sqrt{\frac{5}{4}}$

Alternatively award for substituting $x = \frac{1}{32}$ into **both** sides and making a connection between the two sides by use of an $=$ or $\approx$.

E.g. $\left(1 + \frac{8}{32}\right)^{\frac{1}{2}} = 1 + 4 \times \frac{1}{32} - 8 \times \left(\frac{1}{32}\right)^2 + 32 \times \left(\frac{1}{32}\right)^3$ following through on their expansion

Also implied by $\frac{\sqrt{5}}{2} = \frac{1145}{1024}$ for a correct expansion

It is not enough to state substitute $x = \frac{1}{32}$ into "the expansion" or just the rhs " $1 + 4x - 8x^2 + 32x^3$ "

A1ft: Requires a full (and correct) **explanation** as to how the expansion can be used to estimate $\sqrt{5}$

E.g. Calculates $1 + 4 \times \frac{1}{32} - 8 \times \left(\frac{1}{32}\right)^2 + 32 \times \left(\frac{1}{32}\right)^3$ and multiplies by 2.

This can be scored from an incorrect binomial expansion or a binomial expansion with more terms.

The explanation could be mathematical. So $\frac{\sqrt{5}}{2} = \frac{1145}{1024} \rightarrow \sqrt{5} = \frac{1145}{512}$ is acceptable.

SC: For 1 mark, M1, A0 score for a statement such as "substitute $x = \frac{1}{32}$ into both sides of part (a) and make $\sqrt{5}$ the subject"

Q6.

Question	Scheme	Marks	AOs
(a)	$\sqrt{4-x} = 2\left(1 - \frac{1}{4}x\right)^{\frac{1}{2}}$	M1	2.1
	$\left(1 - \frac{1}{4}x\right)^{\frac{1}{2}} = 1 + \frac{1}{2}\left(-\frac{1}{4}x\right) + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)}{2!}\left(-\frac{1}{4}x\right)^2 + \dots$	M1	1.1b
	$\sqrt{4-x} = 2\left(1 - \frac{1}{8}x - \frac{1}{128}x^2 + \dots\right)$	A1	1.1b
	$\sqrt{4-x} = 2 - \frac{1}{4}x - \frac{1}{64}x^2 + \dots$ and $k = -\frac{1}{64}$	A1	1.1b
		(4)	
(b)	The expansion is valid for $ x < 4$, so $x = 1$ can be used	B1	2.4
		(1)	
(5 marks)			
Notes:			
(a)			
M1: Takes out a factor of 4 and writes $\sqrt{4-x} = 2(1 \pm \dots)^{\frac{1}{2}}$			
M1: For an attempt at the binomial expansion with $n = \frac{1}{2}$			
Eg. $(1+ax)^{\frac{1}{2}} = 1 + \frac{1}{2}(ax) + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)}{2!}(ax)^2 + \dots$			
A1: Correct expression inside the bracket $1 - \frac{1}{8}x - \frac{1}{128}x^2 + \dots$ which may be left unsimplified			
A1: $\sqrt{4-x} = 2 - \frac{1}{4}x - \frac{1}{64}x^2 + \dots$ and $k = -\frac{1}{64}$			
(b)			
B1: The expansion is valid for $ x < 4$, so $x = 1$ can be used			

Q7.

Question	Scheme	Marks	AOs
(a)	$\sqrt{\frac{1+4x}{1-x}} = (1+4x)^{0.5} \times (1-x)^{-0.5}$	B1	3.1a
	$(1+4x)^{0.5} = 1 + 0.5 \times (4x) + \frac{0.5 \times -0.5}{2} \times (4x)^2$	M1	1.1b
	$(1-x)^{-0.5} = 1 + (-0.5)(-x) + \frac{(-0.5) \times (-1.5)}{2} (-x)^2$	M1	1.1b
	$(1+4x)^{0.5} = 1 + 2x - 2x^2$ and $(1-x)^{-0.5} = 1 + 0.5x + 0.375x^2$ oe	A1	1.1b
	$(1+4x)^{0.5} \times (1-x)^{-0.5} = (1+2x-2x^2 \dots) \times \left(1 + \frac{1}{2}x + \frac{3}{8}x^2 \dots\right)$ $= 1 + \frac{1}{2}x + \frac{3}{8}x^2 + 2x + x^2 - 2x^2 + \dots$ $= A + Bx + Cx^2$	dM1	2.1
	$= 1 + \frac{5}{2}x - \frac{5}{8}x^2 \dots *$	A1*	1.1b
		(6)	
(b)	Expression is valid $ x < \frac{1}{4}$ Should not use $x = \frac{1}{2}$ as $\frac{1}{2} > \frac{1}{4}$	B1	2.3
		(1)	
(c)	Substitutes $x = \frac{1}{11}$ into $\sqrt{\frac{1+4x}{1-x}} \approx 1 + \frac{5}{2}x - \frac{5}{8}x^2$	M1	1.1b
	$\sqrt{\frac{3}{2}} = \frac{1183}{968}$	A1	1.1b
	(so $\sqrt{6}$ is) $\frac{1183}{484}$ or $\frac{2904}{1183}$	A1	2.1
		(3)	
(10 marks)			

(a) B1: Scored for key step in setting up the process so that it can be attempted using binomial expansions This could be achieved by $\sqrt{\frac{1+4x}{1-x}} = (1+4x)^{0.5} \times (1-x)^{-0.5}$ See end for other alternatives It may be implied by later work. M1: Award for an attempt at the binomial expansion $(1+4x)^{0.5} = 1 + 0.5 \times (4x) + \frac{(0.5) \times (-0.5)}{2} \times (4x)^2$ There must be three (or more terms). Allow a missing bracket on the $(4x)^2$ and a sign slip so the correct application may be implied by $1 + 2x \pm 0.5x^2$ M1: Award for an attempt at the binomial expansion $(1-x)^{-0.5} = 1 + (-0.5)(-x) + \frac{(-0.5) \times (-1.5)}{2} (-x)^2$ There must be three (or more terms). Allow a missing bracket on the $(-x)^2$ and a sign slips so the method may be awarded on $1 \pm 0.5x \pm 0.375x^2$ A1: Both correct and simplified. This may be awarded for a correct final answer if a candidate does all their simplification at the end dM1: In the main scheme it is for multiplying their two expansions to reach a quadratic. It is for the key step in adding 'six' terms to produce the quadratic expression. Higher power terms may be seen. Condone slips on	
---	--

the multiplication on one term only. It is dependent upon having scored the first B and one of the other two M's

In the alternative it is for multiplying $\left(1 + \frac{5}{2}x - \frac{5}{8}x^2\right)(1-x)^{0.5}$ and comparing it to $(1+4x)^{0.5}$

It is for the key step in adding 'six' terms to produce the quadratic expression.

A1*: Completes proof with no errors or omissions. In the alternative there must be some reference to the fact that both sides are equal.

(b)

B1: States that the expansion may not / is not valid when $|x| > \frac{1}{4}$

This may be implied by a statement such as $\frac{1}{2} > \frac{1}{4}$ or stating that the expansion is only valid when $|x| < \frac{1}{4}$

Condone, for this mark a candidate who substitutes $x = \frac{1}{2}$ into the $4x$ and states it is not valid as $2 > 1$ or

Don't award for candidates who state that $\frac{1}{2}$ is too big without any reference to the validity of the expansion.

As a rule you should see some reference to $\frac{1}{4}$ or $4x$

(c)(i)

M1: Substitutes $x = \frac{1}{11}$ into BOTH sides $\sqrt{\frac{1+4x}{1-x}} \approx 1 + \frac{5}{2}x - \frac{5}{8}x^2$ and attempts to find at least one side.

As the left hand side is $\frac{\sqrt{6}}{2}$ they may multiply by 2 first which is acceptable

A1: Finds both sides leading to a correct equation/statement $\sqrt{\frac{15}{10}} = \frac{1183}{968}$ or $\sqrt{6} = 2 \times \frac{1183}{968}$

A1: $\sqrt{6} = \frac{1183}{484}$ or $\sqrt{6} = \frac{2904}{1183}$ $\sqrt{6} = 2 \times \frac{1183}{968} = \frac{1183}{484}$ would imply all 3 marks

Watch for other equally valid alternatives for 11(a) including

B1: $(1+4x)^{0.5} \approx \left(1 + \frac{5}{2}x - \frac{5}{8}x^2\right)(1-x)^{0.5}$ then the M's are for $(1+4x)^{0.5}$ and $(1-x)^{0.5}$

M1: $(1-x)^{0.5} = 1 + (0.5)(-x) + \frac{(0.5) \times (-0.5)}{2}(-x)^2$

Or

B1: $\sqrt{\frac{1+4x}{1-x}} = \sqrt{1 + \frac{5x}{1-x}} = \left(1 + 5x(1-x)^{-1}\right)^{\frac{1}{2}}$ then the first M1 for one application of binomial and the second would be for both $(1-x)^{-1}$ and $(1-x)^{-2}$

Or

B1: $\sqrt{\frac{1+4x}{1-x}} \times \frac{\sqrt{1-x}}{\sqrt{1-x}} = \sqrt{(1+3x-4x^2)} \times (1-x)^{-1} = \left(1 + (3x-4x^2)\right)^{\frac{1}{2}} \times (1-x)^{-1}$